

Supplemental Material

To split or not to split: Characterizing chemical pollution impacts in aquatic ecosystems with species sensitivity distributions for specific taxonomic groups

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The supplemental material consists of five sections with two supplemental figures and two supplemental tables.

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Additional Files - Excel Document

Experimental test data curation and harmonization

Table A1 Process flow for developing SSD-based hazard concentrations (HC20) for freshwater aquatic species for use in Life Cycle Impact Assessment (LCIA)

#	Step	Description/Explanation
1	Pre-processing	
1a	A unique list of chemicals	- Identify the unique list of chemicals from the list published in Posthuma et al. 2019, and map the list of chemicals for which experimental data were available in the curated raw data underlying Posthuma et al. 2019 - Removal of double entries based on CAS RN search in CompTox Dashboard - Manual identification of validity of CAS RN not listed in CompTox Dashboard - Search missing chemical descriptors from CompTox in chemidplus and pubchem databases
1b	Mapping of chemical classification on the unique list of chemicals	- Extract from ClassyFire chemical taxonomy kingdom, superclass, class, and subclass (where available) for all unique chemicals based on CAS and SMILES matches - Extract from Pesticide Compendium (earlier: Alan Wood) chemical class and target class (e.g., herbicides) for all unique chemicals based on CAS matches - Manual identification of chemical taxonomy for chemicals that were not identifiable in ClassyFire
1c	A unique list of toxic modes of action and mapping on the unique list of chemicals	- Identify from Pesticide Resistance Networks (FRAC, HRAC, IRAC) a unique list of specific modes of action, and extract related chemical groups and example chemicals listed - Extract from Verhaar scheme 4 generic modes of action classes for all unique chemicals based on CAS matches - Map the toxic modes of action on the unique chemicals list
1d	A unique list of species	- Identify the unique list of species from the list available in curated raw data underlying Posthuma et al. 2019, table "Selected_ToX_data_measured" - Correction of misspelled species names based on searching species names in Global Names Resolver - Remove double entries based on genus and species names - Remove varieties and sub-species names and merge those into data for the same species - Replace common species names with scientific species names based on IUCN species lists, and complement by manual searches for common names not listed in IUCN - Replace old and synonym species names with current species names according to IUCN and GBIF taxonomy, and in case of mismatches between IUCN and GBIF, take IUCN as a priority - Extract a list of habitats from the list in curated raw data underlying Posthuma et al. 2019, and assign either "freshwater," "marine," "terrestrial," or "other" as habitat type, and map habitat types on the list of species - Filter out all species that are not associated with "freshwater" as habitat type - Define default species for different taxonomic levels (e.g., family) based on common species used in risk assessment or else based on dominant species within a given level (e.g., genus)

#	Step	Description/Explanation
		- Assign scientific species name to all entries that are reported for higher level only (e.g., family or genus): for dominant species within a given level (e.g., genus), assign this species, else assign no species - Assign artificial species name for levels where no species name is reported, and no default species name can be assigned, based on test-ID in curated raw data underlying Posthuma et al. 2019 (to keep data points separate for supposedly distinct species)
1e	Mapping of taxonomic classification on the unique list of species	- Extract as background list of unique species and their taxonomy (i.e., kingdom, phylum, family, order, class, genus, species) from IUCN (based on web-search function) and from GBIF (based on code-based downloads) - Extract from IUCN, GBIF, or EnviroTox taxonomy databases habitat type (freshwater or saltwater) for all unique species names based on scientific species name matches - Define four main taxonomic groups (i.e., algae, cyanobacteria, and aquatic plants; vertebrates; invertebrates; other microorganisms) in line with "biological quality elements" defined under the EU Water Framework Directive, Annex V (phytoplankton, aquatic flora, benthic invertebrates, and fish fauna) as well as with the concept of trophic levels in aquatic ecosystems (primary producers/autotrophs, primary consumers/ herbivores, secondary consumers/carnivores, decomposers/detritivores): - Algae, cyanobacteria, and aquatic plants, - Invertebrates (crustaceans and non-crustaceans), - Vertebrates (including fish), - Other microorganisms (e.g., bacteria) - Extract from IUCN, GBIF, and EnviroTox lists the taxonomy, habitat type, and main taxonomic group for all unique species names in our dataset based on scientific species name matches - Manually assign taxonomy, habitat type, and main taxonomic group for species names not included in IUCN, GBIF, or EnviroTox based on additional sources (e.g., WoRMS)
1f	A unique list of effect types & exposure test duration and mapping on curated raw data	- Extract from column "TypeToxData" and column "HarmonizedExposureDuration.d." in curated raw data underlying Posthuma et al. 2019, table "Selected_Tox_data_measured," respectively a list of unique effect types and the exposure test durations (in days) - Aggregate effect types into equivalents of NOEC, EC10, and EC50, and remove ambiguous endpoints that cannot be aggregated into one of the given effect types - Assign aggregated effect type and exposure test duration type to all data points in curated raw data underlying Posthuma et al. 2019 - Extract from Table 1 of Aurisano et al. 2019 exposure test duration thresholds (separating test durations into exposure test duration types' acute' or 'chronic') per main taxonomic group - Assign exposure test duration types to reported harmonized exposure test durations of all data points in curated raw data underlying Posthuma et al. 2019, based on the main taxonomic group and exposure test duration matching
2	Dataset preparation	
2a	Extrapolation to chronic EC10 equivalents (EC10^{eq})	The approach used here for consistency - Extract different endpoints test data (i.e., acute NOECs, chronic NOECs, acute EC10, acute EC50, and chronic EC50) - Compare in a separate regression equation against the chronic EC10 for the same species-chemical combination

#	Step	Description/Explanation
		• Use the full regression statistics (e.g., $\text{chronic EC10} = \alpha \times \text{acute NOEC} + \beta$) if the slope is not close to unity • Compare the resulting regression results with Aurisano et al. 2019 Alternative approach • Extract from Aurisano et al. 2019 extrapolation factors from effect types to EC10^{eq} and from exposure test duration types to 'chronic,' separately for each main taxonomic group (Table 3) or, if not available, taxonomic group-generically (Table 4) • Apply the extrapolations factors from Aurisano et al. 2019 to all data points in curated raw data underlying Posthuma et al. 2019 to arrive at a full set of chronic EC10^{eq}
2b	Aggregation of data points per species	• Identify per chemical and exposure test duration type all data points in the full set of chronic EC10^{eq} generated from curated raw data underlying Posthuma et al. 2019 that belong to the same harmonized species • Calculate the arithmetic mean, min-max range, and data point count across all chronic EC10^{eq} data points per chemical-species combination for each exposure test duration type (i.e., once for chronic EC10^{eq} derived only from chronic data and once for chronic EC10^{eq} derived from chronic and acute data)
2c	Criteria definition for a minimum number of data points	• Count for each chemical the number of chronic EC10^{eq} across species per main taxonomic group and the number of main taxonomic groups, separately for each exposure test duration type (i.e., once for chronic EC10^{eq} derived only from chronic data, and once for chronic EC10^{eq} derived from chronic and acute data) • Define as a threshold between data-poor and data-rich chemicals a minimum of 3 distinct species per main taxonomic group (or trophic level) and a minimum of 3 distinct main taxonomic groups (or trophic levels) based on recommendations by Rosenbaum et al. 2008 • Assign the thresholds for data-poor vs. data-rich chemicals to each chemical separately for each exposure test duration type (i.e., once for chronic EC10^{eq} derived only from chronic data and once for chronic EC10^{eq} derived from chronic and acute data) • Flag chemicals that are above the defined thresholds for data-poor vs. data-rich chemicals as 'data-rich,' separately for chemicals only based on chronic data vs. chemicals based on chronic and acute data
3	Effect calculation	
3a	Deriving statistics for single-SSD and split-SSD per chemical	• Convert all chronic EC10^{eq} into logarithmic ($\log_{10}$) scale, i.e. derive a set of $\log_{10}(\text{EC10}^{\text{eq}})$ • Derive the number of $\log_{10}(\text{EC10}^{\text{eq}})$ per chemical (single SSD) and per chemical-main taxonomic group combination (split-SSD per chemical) • Calculate per chemical (i.e., for single SSD) and per chemical-main taxonomic group combination (i.e., for split-SSD) the arithmetic mean and standard deviation of all $\log_{10}(\text{EC10}^{\text{eq}})$, which yields the geometric mean and geometric standard deviation across respective EC10^{eq}
3b	Plotting EC10^{eq} data points on SSD graphs	• Derive the number of $\log_{10}(\text{EC10}^{\text{eq}})$ per chemical • Rank in ascending order each $\log_{10}(\text{EC10}^{\text{eq}})$ per chemical against the count of all $\log_{10}(\text{EC10}^{\text{eq}})$ for the same chemical, e.g. using in Microsoft Excel the RANK.EQ function as 'RANK.EQ($\log_{10}(\text{EC10}^{\text{eq}}$) count, $\log_{10}(\text{EC10}^{\text{eq}}$ value, 1)' • Derive per chemical for each $\log_{10}(\text{EC10}^{\text{eq}})$ as x-value on a scatter plot the corresponding response probability (potentially-affected fraction PAF of species) as y-axis, as 'PAF = ($\log_{10}(\text{EC10}^{\text{eq}}$) rank – H) /

#	Step	Description/Explanation
		$\log_{10}(\text{EC}_{10^{\text{eq}}})$ count', with the Hazen constant $H = 0.5$ (see Barnett 1975)
3c	Plotting fitted SSD graphs	- Define per chemical, and per chemical-main taxonomic group combination the range of $\log_{10}(\text{EC}_{10^{\text{eq}}})$ for the x-axis of a scatter plot with smooth lines - Derive a fitted cumulative normal distribution of $\log_{10}(\text{EC}_{10^{\text{eq}}})$ values (representing a log-normal distribution of $\text{EC}_{10^{\text{eq}}}$) over the specified x-axis range, e.g., using in Microsoft Excel the NORM.DIST function as 'NORM.DIST($\log_{10}(\text{EC}_{10^{\text{eq}}})$ value, arithmetic mean, standard deviation, 1)', as detailed in, e.g., Chapter 5 of Posthuma et al. 2002
3d	Deriving HC20 per SSD	- Define on an SSD graph the value 0.2 as the percentile at which 20% of species per chemical or per chemical-main taxonomic group combination show a response (y-axis) at their respective $\log_{10}(\text{EC}_{10^{\text{eq}}})$ values (x-axis) - Define this percentile as HC20 on each SSD, denoting the hazard concentration at which 20% of species show a probable effect above their $\log_{10}(\text{EC}_{10^{\text{eq}}})$; see Owsianiak et al. 2019 - Calculate the $\log(\text{HC20})$ from the fitted SSD as the inverse of the normal cumulative distribution, e.g., using Microsoft Excel the NORM.INV function as 'NORM.INV(0.2, arithmetic mean, standard deviation)' - Derive the final HC20 as '$\text{HC20} = 10^{\log(\text{HC20})}$', as detailed in, e.g., Chapter 5 of Posthuma et al. 2002
3e	Comparison of HC20	- Define for split-SSD a set of combined criteria for indicating significantly different SSDs across chemical-main taxonomic group combinations: - Fitted SSD curves are not crossing each other over the relevant $\log_{10}(\text{EC}_{10^{\text{eq}}})$ range, 95% confidence intervals of SSD curves are not overlapping over the relevant $\log_{10}(\text{EC}_{10^{\text{eq}}})$ range, or SSD-related HC20 values show at least a factor of 10 difference - The number of data points is high enough to build a robust SSD (i.e., a minimum of 10 data points representing distinct species within one main taxonomic group or a minimum of 10 data points and at least 3 data points per main taxonomic group) - Label split-SSDs as either significantly different or not based on applying the set of combined criteria - Merge split-SSDs that are not significantly different into a combined SSD and redo the comparison with other split-SSDs, yielding ultimately a set of statistically significant split-SSDs per chemical, indicating main taxonomic groups that are specifically targeted (i.e., related species mainly showing effects at lower $\text{EC}_{10^{\text{eq}}}$) as compared to main taxonomic groups that are not targeted (i.e., related species mainly showing effects at higher $\text{EC}_{10^{\text{eq}}}$) - Label those chemicals for which at least one main taxonomic group fulfills the criteria for a statistically significant split-SSD
3f	Statistical tests for evaluating the potential of splitting SSD	- Follow the statistical tests steps in the systematic evaluation framework for splitting SSDs, as shown in Figure 1 - Evaluate whether to split SSD or not based on the statistical outcome with the MoA information, use category, visually check the SSD 95% confidence interval - Responsible split has low uncertainty around the HC20, no overlapping or crossing CI, and has a different level of sensitivity (HC20) from the rest of the group

#	Step	Description/Explanation
4	Uncertainty assessment	
5a	Preliminary steps	• Determine σ for each combination (chemical-species-effect type) to analyse how sd varies to set up a fixed sd in case of taxonomic group combination with a low number of data points (fixed shaped SSD)
5b	Total uncertainty around the derived HC20	• Quantifying uncertainty around the derived HC20 values by combining two types of uncertainty: GSD_{inter}^2 reflecting inter-species variability (i.e., variability across available effect values), and GSD_{intra}^2 reflecting intra-species variability (i.e., variability around the effect values). More details are in Table A2

Uncertainty analysis

Table A2 presents the specific steps to quantify the total uncertainty around the calculated HC20 values; GSD_{total}^2 (squared geometric standard deviation) combining both the GSD_{inter}^2 inter-species variability and GSD_{intra}^2 intra-species variability per taxonomic group-chemical combination (for simplicity, from now on referred to as 'combination') is described in more detail in the following section.

Table A2 Overview of the approach for quantifying uncertainty around the derived HC20 values by combining two types of uncertainty: GSD_{inter}^2 reflecting inter-species variability (i.e., variability across available effect values), and GSD_{intra}^2 reflecting intra-species variability (i.e., variability around the effect values).

Inter-species variability We calculated GSD_{inter}^2, which reflects the variability across available effect values for each taxonomic group per chemical combination, hereafter referred to as combination HC20. To estimate GSD_{inter}^2 we started from the lognormal distribution fitted through the available effect values extrapolated chronic EC10 equivalent. When fitting the lognormal distribution, one of the two moments used is σ (standard deviation of the available for a chemical). We thus estimated the 95% <i>CI</i> (confidence interval) of σ via the function "fitdistr" in the R package MASS¹, and from this 95% <i>CI</i> we derived an upper and lower bound for $HC20_{chronic\ EC10\ equivalent}$ ($HC20_{chronic\ EC10\ equivalent}^{inter,upper}$ and $HC20_{chronic\ EC10\ equivalent}^{inter,lower}$) by fitting two new lognormal distributions using instead of σ its 95% <i>CI</i>. GSD_{inter}^2 is then calculated as:^{2,3} $GSD_{inter}^2 = \sqrt{HC20_{chronic\ EC10\ equivalent}^{inter,upper} / HC20_{chronic\ EC10\ equivalent}^{inter,lower}}$ We did not derive a chemical-specific group combination GSD_{inter}^2 in the case of the data-poor taxonomic group (< 6 records available) because GSD_{inter}^2 might be highly biased by the limited number of effect values available. For these chemicals, we applied a fixed GSD_{inter}^2 calculated as 97.5 %-ile of the estimated GSD_{inter}^2 across taxonomic groups with six records available.
Intra-species variability We calculated GSD_{intra}^2, which reflects variability specific to the effect values for each combination HC20_{chronic EC10 equivalent}. To estimate GSD_{intra}^2 we started from the record-specific distribution around the extrapolated effect value. This record-specific

distribution is based on the uncertainty distributions assigned when extrapolating the *measured* data to chronic EC10 equivalent from different endpoints (NOEC, EC50, and EC10). The record-specific uncertainty is propagated from the extrapolated chronic EC10 equivalent to the derived $HC20_{\text{chronic EC10 equivalent}}$ via a bootstrap method. Firstly, 1000 bootstrap samples were sampled from the estimated distributions around the extrapolated chronic EC10 equivalent for each combination. Secondly, 1000 lognormal distributions were fitted to the bootstrap samples using μ as the median of the resampled effect values and σ as the same σ used to derive $HC20_{\text{chronic EC10 equivalent}}$, based on the originally available effect values (in practice, only μ varies, and always the same shaped distribution is fitted to the resamples). Thirdly, from the 1000 fits, we derived an upper and lower bound for $HC20_{\text{chronic EC10 equivalent}}$ ($HC20_{\text{chronic EC10 equivalent}}^{\text{intra,upper}}$ and $HC20_{\text{chronic EC10 equivalent}}^{\text{intra,lower}}$). GSD_{intra}^2 was then calculated as:^{2,3}

$$GSD_{\text{intra}}^2 = \sqrt{HC20_{\text{chronic EC10 equivalent}}^{\text{intra,upper}} / HC20_{\text{chronic EC10 equivalent}}^{\text{intra,lower}}}$$

We could calculate the upper and lower bounds for all the combinations since we had at least 3 unique species per taxonomic group.

Total Uncertainty

To determine the total uncertainty in $HC20_{\text{chronic EC10 equivalent}}$ we combined the resulting inter-species variability with that resulting from the intra-species variability:

$$GSD_{\text{total}}^2 = 10^{\sqrt{\log_{10}((GSD_{\text{inter}}^2)) + \log_{10}((GSD_{\text{intra}}^2))}}$$

The calculated $HC20$ effect threshold per taxonomic group and its 95% CI account for the inter- and intra-species variability.

SSDs robustness check from total uncertainty

To assess robust SSDs to determine "responsible split" SSDs, preferred full split, partial split as the fallback option, and the overall non-robust SSDs. We pragmatically set the GSD_{total}^2 of 5 (i.e., less than 5 orders of magnitude) as the criterion to define sufficiently robust SSDs, where SSDs were considered robust when statistically shown to be significantly different from the rest and uncertainty is ≤ 5 .

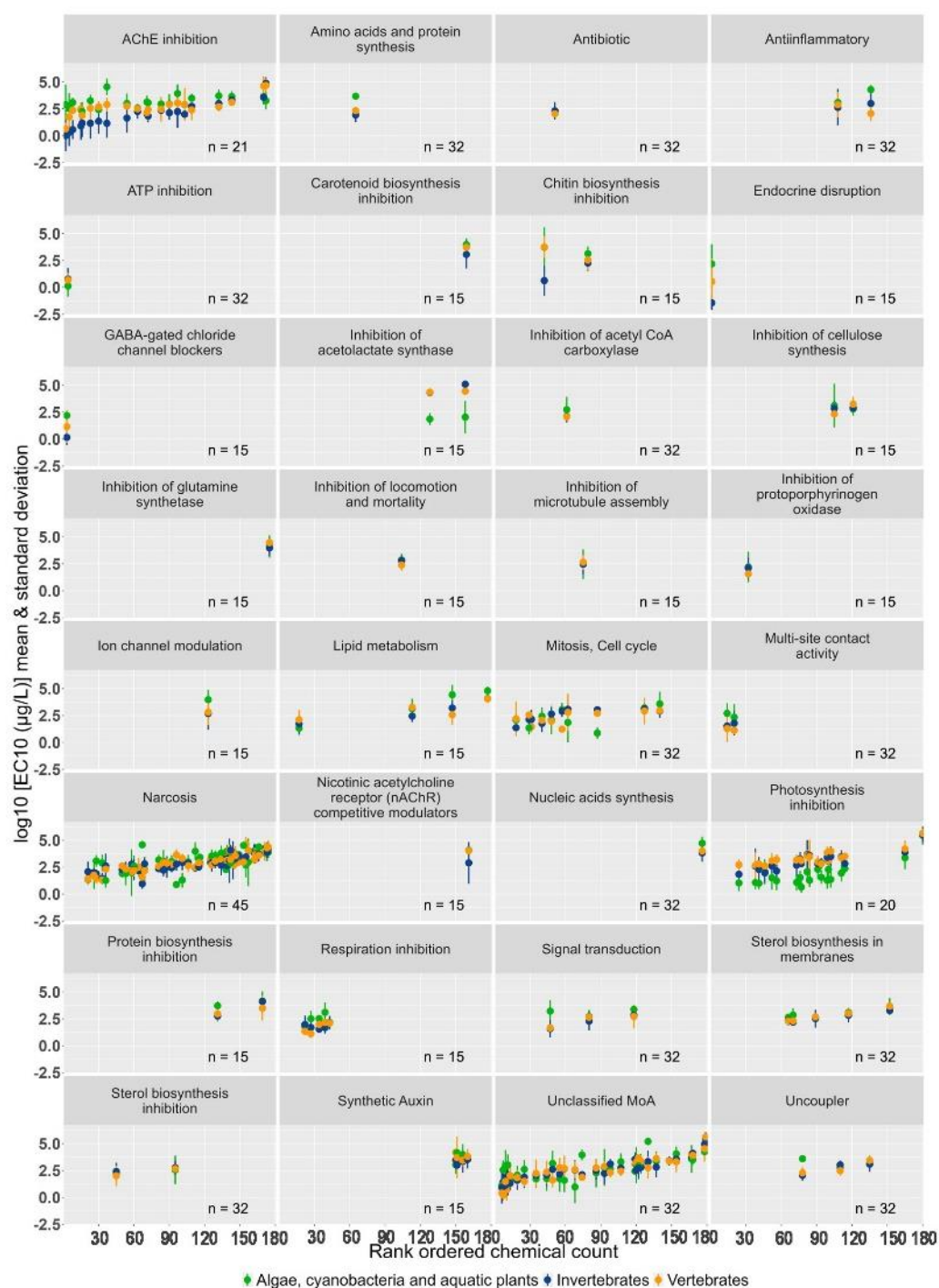


Figure S1 Distributions of chronic EC10-equivalent endpoint data for 180 chemicals, summarized as taxonomic group means and standard deviations, and considering different mechanistic modes of action (MoA). Chemicals in each plot are rank ordered based on the mean values calculated across all data per chemical. SSD-splitting grossly coincides (visually) with non-overlapping dots and standard deviations for different taxonomic groups (colors).

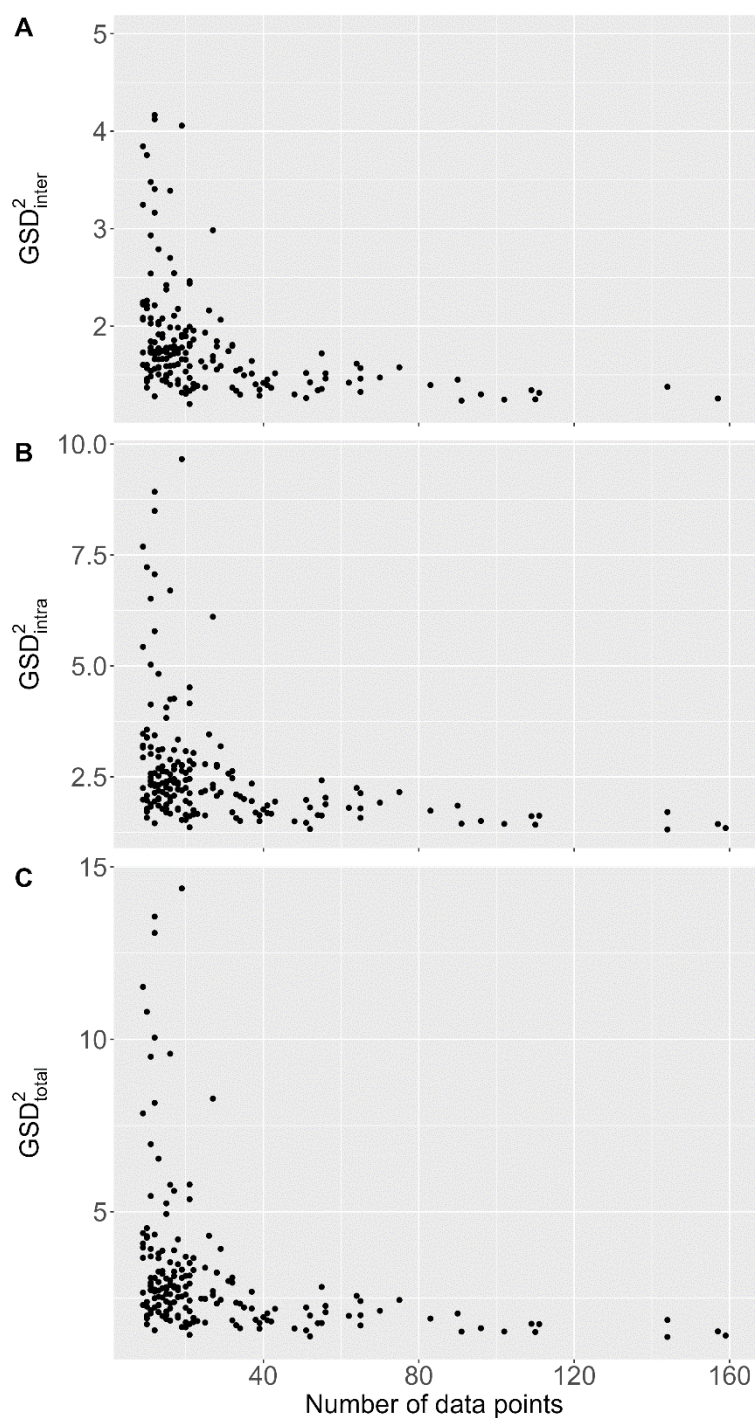


Figure S2 A. $\text{GSD}_{\text{inter}}^2$ (inter-species variability), **B.** $\text{GSD}_{\text{intra}}^2$ (intra-species variability) and **C.** $\text{GSD}_{\text{total}}^2$ (total uncertainty) as a function of the number of data points available for each taxonomic group chemical combination. $\text{GSD}_{\text{total}}^2$ combines both the $\text{GSD}_{\text{inter}}^2$ and $\text{GSD}_{\text{intra}}^2$.

R code for split SSDs statistical analysis

```
###Library
library(readxl)
library(tidyverse)
library(broom)
library(car)
library(FSA)

#Import Excel Table S1
ecotox_data_CAS <- read_excel("... /Supplemental Excel File.xlsx", sheet = "Excel Table
S1", skip = 2)

## Split SSDs test
Step A: Check for Homogeneity of variance https://datatab.net/tutorial/levене-test
Step B: ANOVA test if Levene output is  $p > 0.05$ 
    B1. Do a post-hoc test -Tukey's HSD- post-hoc test for pairwise comparisons- to check
    which group is different only if anova  $p \leq 0.05$  (significant)
    B2. Compare two groups e.g A versus I using independent t test and A vs V+I --if Anova
     $p \leq 0.05$ 
Step C: Kruskal-Wallis test if Levene output is  $p < 0.05$  (Non parametric equivalent of
ANOVA)
    C1. Do a post-hoc test (Dunn test- post-hoc test for pairwise comparisons- to check which
    group is different only if Kruskal Wallis  $p \leq 0.05$  (significant)
    C2. Compare two groups e.g A versus I using Mann Whitney U t test and A versus V+I --
    if Kruskal Wallis  $p \leq 0.05$ 
Step f: Linear regression with species groups as independent variables e.g A vs I and A vs
V+I --if Kruskal Wallis  $p \leq 0.05$ 

### shorthand species group names
ecotox_data_CAS$renamed_speciesgroup <- case_when(
  ecotox_data_CAS$SpeciesGroup_Harmonized=="Algae, cyanobacteria and aquatic plants"
~ "A",
  ecotox_data_CAS$SpeciesGroup_Harmonized=="Invertebrates" ~ "I",
  .default = "V"
)

ecotox_data_CAS$SpeciesGroup_Harmonized.fac <-
factor(ecotox_data_CAS$renamed_speciesgroup)
#### merged groups
#For independent test-comparing means between 1 group vs two other groups

ecotox_data_CAS$merged_VI <-factor(ifelse(ecotox_data_CAS$renamed_speciesgroup
%in% c("V", "I"), "VI", "A"))
ecotox_data_CAS$merged_VA <-factor(ifelse(ecotox_data_CAS$renamed_speciesgroup
%in% c("V", "A"), "VA", "I"))
ecotox_data_CAS$merged_IA <-factor(ifelse(ecotox_data_CAS$renamed_speciesgroup
%in% c("I", "A"), "IA", "V"))

## Make a list to store the results
```

```
CASSPLIT <- aggregate(SpeciesName_Harmonized~CAS_Harmonized, data =
ecotox_data_CAS, FUN = length)
```

```
CASSPLIT$leveneTest <- NA
CASSPLIT$anovaTest <- NA
CASSPLIT$TukeyTest_I.A<-NA
CASSPLIT$TukeyTest_V.A<-NA
CASSPLIT$TukeyTest_V.I<-NA
CASSPLIT$Independent_V.IA <- NA
CASSPLIT$Independent_I.VA <- NA
CASSPLIT$Independent_A.VI <- NA
CASSPLIT$KruskalTest <-NA
CASSPLIT$dunnTest_I.A<-NA
CASSPLIT$dunnTest_V.A<-NA
CASSPLIT$dunnTest_V.I<-NA
CASSPLIT$ManW_V.IA <-NA
CASSPLIT$ManW_I.VA <-NA
CASSPLIT$ManW_A.VI <-NA
CASSPLIT$model_V.IA <-NA
CASSPLIT$model_I.VA <-NA
CASSPLIT$model_A.VI <-NA
CASSPLIT$model_V.IA_QC<-NA ##checking if the right p values are extracted
CASSPLIT$model_V.I <-NA
CASSPLIT$model_A.V <-NA
CASSPLIT$model_A.I <-NA
```

```
## for loop comparing means
```

```
for(irow in 1:nrow(CASSPLIT)) { #irow = 1
  iCAS <- CASSPLIT$CAS_Harmonized[[irow]]
  data <-ecotox_data_CAS %>%
    filter(CAS_Harmonized == iCAS)

  varleveneTest <- leveneTest(log.EC10 ~ SpeciesGroup_Harmonized.fac, data=data)
  CASSPLIT$leveneTest[[irow]] <- varleveneTest$`Pr(>F)`[[1]]

  if(varleveneTest$`Pr(>F)`[[1]] > 0.05){

    varanovaTest <- aov(log.EC10 ~ SpeciesGroup_Harmonized.fac, data = data)
    sum_test<-unlist(summary(varanovaTest))
    CASSPLIT$anovaTest[[irow]] <- sum_test[["Pr(>F)1"]]

    if(sum_test[["Pr(>F)1"]] <= 0.05){
      varTurkeyTest<-TukeyHSD(varanovaTest, conf.level=.95)
      TukeyHSD_result <- varTurkeyTest[["SpeciesGroup_Harmonized.fac"]]
      CASSPLIT$TukeyTest_I.A[[irow]]<-TukeyHSD_result[["I-A","p adj"]]
      CASSPLIT$TukeyTest_V.A[[irow]]<-TukeyHSD_result[["V-A","p adj"]]
      CASSPLIT$TukeyTest_V.I[[irow]]<-TukeyHSD_result[["V-I","p adj"]]
    }
  }
}
```

```

}

if(sum_test[["Pr(>F)1"]] <= 0.05){
  varIndependentTest_VI <- t.test(log.EC10 ~ merged_VI, data=data)
  CASSPLIT$Independent_A.VI[[irow]]<-
  varIndependentTest_VI[["p.value"]]
}

if(sum_test[["Pr(>F)1"]] <= 0.05){
  varIndependentTest_VA <- t.test(log.EC10 ~ merged_VA, data=data)
  CASSPLIT$Independent_I.VA[[irow]]<-
  varIndependentTest_VA[["p.value"]]
}

if(sum_test[["Pr(>F)1"]] <= 0.05){
  varIndependentTest_IA <- t.test(log.EC10 ~ merged_IA, data=data)
  CASSPLIT$Independent_V.IA[[irow]]<-
  varIndependentTest_IA[["p.value"]]
}

}

if(varleveneTest$`Pr(>F)`[[1]] < 0.05){
  varKruskalTest <- kruskal.test(log.EC10 ~ SpeciesGroup_Harmonized.fac,
                                data = data)
  CASSPLIT$KruskalTest[[irow]] <- varKruskalTest[["p.value"]]

  if(varKruskalTest[["p.value"]] <= 0.05){
    dunnTest_result<-
    dunnTest(log.EC10 ~ SpeciesGroup_Harmonized.fac,
             data=data,method="bonferroni")
    #dunn_result <- dunnTest_result[["res"]]
    CASSPLIT$dunnTest_I.A[[irow]]<-
    dunnTest_result[["res"]][["P.adj"]][1]

    CASSPLIT$dunnTest_V.A[[irow]] <-
    dunnTest_result[["res"]][["P.adj"]][2]

    CASSPLIT$dunnTest_V.I[[irow]] <-
    dunnTest_result[["res"]][["P.adj"]][3]

  }

  if(varKruskalTest[["p.value"]] <= 0.05){
    varManW_VA <- tryCatch(

```

```

    wilcox.test(log.EC10 ~ merged_VA, data=data),
    error = stop,
    warning = function(cond){
      return(list(p.value = 0)) #or use jitter to suppress?
    })
  CASSPLIT$ManW_I.VA[[irow]]<-
    varManW_VA[["p.value"]]
}

if(varKruskalTest[["p.value"]] <= 0.05){
  varManW_IA <-
    wilcox.test(log.EC10 ~ merged_IA, data=data)
  CASSPLIT$ManW_V.IA[[irow]]<-
    varManW_IA[["p.value"]]
}
if(varKruskalTest[["p.value"]] <= 0.05){
  varManW_VI <-
    wilcox.test(log.EC10 ~ merged_VI, data=data)
  CASSPLIT$ManW_A.VI[[irow]]<-
    varManW_VI[["p.value"]]

}
}

}

#NA's drop out -> test is not performed
cat("levene.test, equal variances p> 0.05")
table(CASSPLIT$leveneTest > 0.05)
cat("for levene.test p > 0.05 == FALSE")
cat("KruskalTest, differences in mean")
table(CASSPLIT$KruskalTest <= 0.05)
cat("dunnTest, differences in mean, 3way")
table(CASSPLIT$dunnTest_I.A <= 0.05 | CASSPLIT$dunnTest_V.A <= 0.05 |
CASSPLIT$dunnTest_V.I <= 0.05 )
cat("Mann Whitney, differences in mean, 3way")
table(CASSPLIT$ManW_A.VI <= 0.05 | CASSPLIT$ManW_I.VA <= 0.05 |
CASSPLIT$ManW_A.VI <= 0.05 )
cat("full?")
table((CASSPLIT$ManW_A.VI <= 0.05) + (CASSPLIT$ManW_I.VA <= 0.05) +
(CASSPLIT$ManW_A.VI <= 0.05) == 3 )

cat("for levene.test p > 0.05 == TRUE")
cat("One way ANOVA, differences in mean")
table(CASSPLIT$anovaTest <= 0.05)
cat("TukeyTest, differences in mean, 3way")
table(CASSPLIT$TukeyTest_I.A <= 0.05 | CASSPLIT$TukeyTest_V.A <= 0.05 |
CASSPLIT$TukeyTest_V.I <= 0.05 )
cat("Independent test, 3way")
table(CASSPLIT$Independent_V.IA <= 0.05 | CASSPLIT$Independent_I.VA <= 0.05 |
CASSPLIT$Independent_A.VI <= 0.05 )

```

```
cat("full?")
table((CASSPLIT$Independent_V.IA <= 0.05) + (CASSPLIT$Independent_I.VA <= 0.05) +
(CASSPLIT$Independent_A.VI <= 0.05) == 3)
```

###comparing models; 1 group vs 2 other groups

```
for(irow in 1:nrow(CASSPLIT)) {
  iCAS <- CASSPLIT$CAS_Harmonized[[irow]]
  data <- ecotox_data_CAS %>%
    filter(CAS_Harmonized == iCAS)
  model_IA <- lm(log.EC10 ~ merged_IA, data = data)

  tidy_model_IA <- tidy(model_IA)
  CASSPLIT$model_V.IA_QC[[irow]] <-
    unlist(tidy_model_IA[term == "merged_IaV", "p.value"])
  #yes, it's the same

  CASSPLIT$model_V.IA[[irow]] <-
    summary(model_IA)$coefficients[, "Pr(>|t)"][[2]]
}
```

```
for(irow in 1:nrow(CASSPLIT)) {
  iCAS <- CASSPLIT$CAS_Harmonized[[irow]]
  data <-ecotox_data_CAS %>%
    filter(CAS_Harmonized == iCAS)
  model_VA <- lm(log.EC10 ~ merged_VA, data=data)
  CASSPLIT$model_I.VA[[irow]]<-
    summary(model_VA)$coefficients[, "Pr(>|t)"][[2]]
}
```

```
for(irow in 1:nrow(CASSPLIT)) {
  iCAS <- CASSPLIT$CAS_Harmonized[[irow]]
  data <- ecotox_data_CAS %>%
    filter(CAS_Harmonized == iCAS)

  model_VI <- lm(log.EC10 ~ merged_VI, data=data)

  CASSPLIT$model_A.VI[[irow]]<-
    summary(model_VI)$coefficients[, "Pr(>|t)"][[2]]
}
```

###comparing 1 group vs 1 other group regression
##merged group model versus one used as an anchor

```
for(irow in 1:nrow(CASSPLIT)) {
  iCAS <- CASSPLIT$CAS_Harmonized[[irow]]
  data <-ecotox_data_CAS %>%
```

```

        filter(CAS_Harmonized == iCAS &
               merged_VI == "VI" )

if(nrow(data>1)){
  merged_V.I <- lm(log.EC10 ~ SpeciesGroup_Harmonized.fac, data=data)
  CASSPLIT$model_V.I[[irow]]<-
    summary(merged_V.I)$coefficients[, "Pr(>|t|)"][[2]]
}

}

for(irow in 1:nrow(CASSPLIT)) {
  iCAS <- CASSPLIT$CAS_Harmonized[[irow]]
  data <- ecotox_data_CAS %>%
    filter(CAS_Harmonized == iCAS &
           merged_VA == "VA" )
  merged_A.V <- lm(log.EC10 ~ SpeciesGroup_Harmonized.fac, data=data)
  CASSPLIT$model_A.V[[irow]] <-
    summary(merged_A.V)$coefficients[, "Pr(>|t|)"][[2]]
}

}

for(irow in 1:nrow(CASSPLIT)) {
  iCAS <- CASSPLIT$CAS_Harmonized[[irow]]
  data <- ecotox_data_CAS %>%
    filter(CAS_Harmonized == iCAS &
           merged_IA == "IA" )

  merged_A.I <- lm(log.EC10 ~ SpeciesGroup_Harmonized.fac, data=data)

  CASSPLIT$model_A.I[[irow]] <-
    summary(merged_A.I)$coefficients[, "Pr(>|t|)"][[2]]
}

```

R code for deriving split SSDs based on the process flow Experimental test data curation and harmonization

Table A1

```

####Library
library(ggplot2)
library(tidyverse)
library(readr)
library(scales)
library(tidyr)
library(readxl)
library(dplyr)
library(psych)
library(boot)
require(fitdistrplus)

```



```

library(cowplot)

## Input data
##All step for species taxonomy harmonization was done in excel; step 1a to 1e
ecotox_datain <- read_excel("... /Supplemental Excel File.xlsx", sheet = "Excel Table S1",
skip = 2)

####subset required data columns
ecotox_datain <-
ecotox_datain[,c("ID","test_id","CAS_Harmonized","NameHarmonized","Name","SpeciesNa
me_Harmonized", "SpeciesGroup_Harmonized","SpeciesName_Harmonized (distinct
species)","Genus_Harmonized",
"SpeciesGroupExposure_Harmonized","Conc1Harmonized.ug.L.",
"TypeToxData","Endpoint_Harmonized","EffCode",
"endpoint","HarmonizedExposureDuration.d.","ChronicDuration",
"Habitat_Aggregated_Harmonized")]

#### select a chemical/s: Simazine used as an example

selected_CAS<-c("122-34-9")# Simazine

ecotox_data<- ecotox_datain %>%
  filter(Habitat_Aggregated_Harmonized == "Freshwater aquatic",CAS_Harmonized ==
selected_CAS)

ecotox_data %>%
  group_by(CAS_Harmonized) %>%
  slice_head()

# Function for saving and displaying a plot.

import::from(ggplot2, ggsave)

save_and_display <- function(fig, filename, width = 6, height = 4){
  filepath <- paste("Figures/", filename, ".", "png", sep = "")
  ggsave(filepath, fig, width = width, height = height)
  return(fig)
}

####create a path for figures
plot_directory <- "Figures"

unlink(paste(plot_directory, "/*", sep = ""))
dir.create(plot_directory, recursive = TRUE)

##Step1f:Exposure duration
#Exposure duration type allocation- acute or chronic
#We used extrapolation factors from Aurisano et al. 2019

```



```

ecotox_data_extrapolated$EC10 = ifelse(ecotox_data_extrapolated$Harmonised_effect.type
%in% "Acute NOEC",ecotox_data_extrapolated $Conc1Harmonized.ug.L.^0.816/ 0.953,
      ifelse(ecotox_data_extrapolated$Harmonised_effect.type %in%
"Chronic NOEC",ecotox_data_extrapolated $Conc1Harmonized.ug.L.^0.965/1.394,
      ifelse(ecotox_data_extrapolated$Harmonised_effect.type %in%
"Acute EC50",ecotox_data_extrapolated $Conc1Harmonized.ug.L.^0.869/3.219,
      ifelse(ecotox_data_extrapolated$Harmonised_effect.type
%in% "Chronic EC50",ecotox_data_extrapolated $Conc1Harmonized.ug.L.^0.872/0.185,

ifelse(ecotox_data_extrapolated$Harmonised_effect.type %in% "Acute
EC10",ecotox_data_extrapolated $Conc1Harmonized.ug.L.^0.813/0.108,

ifelse(ecotox_data_extrapolated$Harmonised_effect.type %in% "Chronic
EC10",ecotox_data_extrapolated $Conc1Harmonized.ug.L./1,
      NA))))))

```

Aggregation of data points per species

#Step 2: First average or geometric mean for the same effect level (NOEC vs. EC10 vs. EC50) and then conversion to EC10eq?

```

ecotox_data_extrapolated <- ecotox_data_extrapolated %>%
  group_by(CAS_Harmonized, SpeciesName_Harmonized) %>%
  mutate(n = n(),
         EC10_C = mean(EC10,na.rm = TRUE ),
         max_c = max(EC10),
         min_c = min(EC10)) %>%
  ungroup()
####Keeping distinct species values
#Using species-chemical chemical combination

```

```

ecotox_data_extrapolated <- ecotox_data_extrapolated %>%
  group_by(CAS_Harmonized) %>%
  distinct_at('SpeciesName_Harmonized',.keep_all = TRUE)

```

Convert EC10 to logEC10;log and log 10 of values is different in r but same in excel

```

ecotox_data_extrapolated_logged <- ecotox_data_extrapolated %>%
  mutate(log_EC10=log10(EC10_C))

```

##Step2b:Data poor and data rich chemicals

#Set data poor and data rich chemicals, by first summing the total number of species per chemical and species group

```

ecotox_data_extrapolated_logged <-ecotox_data_extrapolated_logged %>%
  group_by(CAS_Harmonized) %>%
  mutate(n_total=n(),
         tax.count = (n_distinct(SpeciesGroup_Harmonized)))

```

```

ecotox_data_extrapolated_logged <-ecotox_data_extrapolated_logged %>%
  group_by(CAS_Harmonized,SpeciesGroup_Harmonized) %>%

```

```

mutate(species.count = n())

####step2c:Identification of data poor chemicals
#Criteria definition for minimum number of data points

ecotox_data_extrapolated_logged <-ecotox_data_extrapolated_logged %>%
  mutate(
    newcol = case_when(
      tax.count < 3 ~ 'poor',
      species.count < 3 ~ 'poor',
      tax.count >= 3 & species.count < 3 ~ 'poor',
      tax.count >= 3 & species.count >= 3 ~ 'rich'

    )
  )

d <- ecotox_data_extrapolated_logged %>% group_by(CAS_Harmonized) %>%
  summarise(Condition = sum(newcol == "poor")) %>% mutate(Adequate.data =
  ifelse(Condition > 0 , "poor","rich"))
d

ecotox_data_extrapolated_logged<- left_join(ecotox_data_extrapolated_logged, d, by=
c("CAS_Harmonized"="CAS_Harmonized"))

#### Overview of the number of data poor and data rich chemicals:select data rich chemicals

poor_richchem <-ecotox_data_extrapolated_logged %>%
  group_by(CAS_Harmonized) %>%
  count(Adequate.data)
poor_richchem

ecotox_data_rich <- filter(ecotox_data_extrapolated_logged,
  Adequate.data=="rich")

## Step3:Effect calculation
##one can use order, arrange or sort to get frac or cumulative concentrations==PAF

ecotox_data_rich_CAS <- ecotox_data_rich %>%
  group_by(CAS_Harmonized) %>%
  arrange((log.EC10), .by_group = TRUE) %>%
  mutate(frac_C=ppoints(log.EC10,0.5)) %>%
  ungroup()

ecotox_data_rich_taxa<- ecotox_data_rich %>%
  group_by(CAS_Harmonized, SpeciesGroup_Harmonized)%>%
  arrange((log.EC10), .by_group = TRUE) %>%
  mutate(frac_t=ppoints(log.EC10,0.5)) %>%
  ungroup()

```

```

## Calculating deterministic HC20
### Calculating deterministic HC20 per chemical

ecotox_data_CAS <- ecotox_data_rich_CAS %>%
  group_by(CAS_Harmonized) %>%
  mutate(n_C = n(),
         mu_C = mean(log.EC10),
         SDV_C = sd(log.EC10),
         lwr=mu_C-SDV_C,
         upr=mu_C+SDV_C) %>%
  ungroup()
ecotox_data_CAS <- ecotox_data_CAS %>%
  mutate(HC20_C = qnorm(0.2, mean=mu_C, sd=SDV_C),
         HC5_C=qnorm(0.05, mean=mu_C, sd=SDV_C))

### Deterministic HC20 per chemical-Speciesgroups
#We want the mean and SD per chemical AND species group and HC20 per species group

ecotox_data_TAXA <-ecotox_data_rich_taxa %>%
  group_by(CAS_Harmonized, SpeciesGroup_Harmonized) %>%
  mutate(n_t = n(),
         mu_t = mean(log.EC10),
         SDV_t = sd(log.EC10),
         lwr_t=mu_t-SDV_t,
         upr_t=mu_t+SDV_t) %>%
  ungroup()

ecotox_data_TAXA <- ecotox_data_TAXA%>%
  mutate(HC20_t = qnorm(0.2, mean=mu_t, sd=SDV_t),
         HC5_t = qnorm(0.05, mean=mu_t, sd=SDV_t))

##Step3
###Plotting species distribution per chemicals

cols <- c( "#00B612" , "#0F4392","#ff9900")
cols.2 <- c( "#0F4392","#575757")
cols.3<- c( "#00B612", "#575757")
cols.4<- c( "#575757","#ff9900")

newxs <- seq(-2.5, 5.0, length.out = 1000)
ecotox_data_CAS$labels <- paste0("n", " = ", ecotox_data_CAS$n_total)

new_label_No_split <- as_labeller(c(`Simazine` = "No split"))

fig0<- ggplot(data = ecotox_data_CAS)+
  geom_point(aes(x=log.EC10, y=frac_C, color = SpeciesGroup_Harmonized,
shape=SpeciesGroup_Harmonized),size=5)+
  scale_color_manual(values = cols)+
  labs(x=expression(paste('log10 [EC10 (µg/L)]')),
       y='Potentially affected species fraction') +
  annotation_logticks(sides='b')+

```

```

facet_wrap(~Name, ncol=1, labeller = new_label_No_split) +
theme(strip.text.x = element_text(size =20))+

geom_text(data=ecotox_data_CAS, aes(1, 0.95,label=labels), size=4)

####Deterministic curve per chemical

d_fit_chemical <- c(unique(ecotox_data_CAS$CAS_Harmonized))
d_fit_chemical <- as.data.frame(d_fit_chemical)
colnames(d_fit_chemical) [1] <- "CAS_Harmonized"

distribution_parameters_C <- ecotox_data_CAS %>%
  group_by(CAS_Harmonized) %>%
  slice(1)

d_fit_chemical <- d_fit_chemical %>%
  left_join(distribution_parameters_C[,c("CAS_Harmonized", "mu_C", "SDV_C", "Name")],
    by=c("CAS_Harmonized")) %>%
  right_join(
    tidyr::expand(d_fit_chemical, CAS_Harmonized, EC = newxs),
    by = c("CAS_Harmonized"))

d_fit_chemical <- d_fit_chemical %>%
  mutate(PAF_C=pnorm(EC, mean=mu_C, sd=SDV_C))
# mutate(PAF_C=pnorm(EC, mean=log(EC10$estimate[1], #sd=log(EC10$estimate[2])))

SSD_chemical <- ggplot()+

  scale_color_manual(values = cols)+
  geom_line(data=d_fit_chemical,
    aes(x=EC,
      y=PAF_C), color="DARK GRAY") +
  geom_smooth() +
  labs(x = expression(paste('log10 [EC10 (µg/L)]')),
    y = 'Potentially affected species fraction')+
  geom_segment(aes(x =ecotox_data_CAS$HC20_C, y = 0.0, xend =
ecotox_data_CAS$HC20_C, yend = 0.2),color = "red", size=0.9)+

  geom_segment(aes(x =ecotox_data_CAS$HC5_C, y = 0.0, xend =
ecotox_data_CAS$HC5_C, yend = 0.05),color = "black", size=0.9)+#1.2

  facet_wrap(~Name, ncol=1, labeller=new_label_No_split) +
  theme(strip.text.x = element_text(size = 20))

#### Species group deterministic curves:Creating plots per chemicals and species group
combination to create deterministic fit

```

```

new_label_full_split <- as_labeller(c(`Simazine` = "Full split"))

d_fit_taxa <- unique(ecotox_data_TAXA[c("CAS_Harmonized")])
d_fit_taxa <- as.data.frame(d_fit_taxa)
colnames(d_fit_taxa)[1] <- c("CAS_Harmonized")

distribution_parameters_t <- ecotox_data_TAXA %>%
  group_by(CAS_Harmonized, SpeciesGroup_Harmonized) %>%
  slice(1)

d_fit_taxa <- d_fit_taxa %>%
  left_join(distribution_parameters_t[,c("CAS_Harmonized", "SpeciesGroup_Harmonized",
    "mu_t", "SDV_t", "Name")],
    by=c("CAS_Harmonized")) %>%
  right_join(
    tidyr::expand(d_fit_taxa, CAS_Harmonized, EC = newxs),
    by = c("CAS_Harmonized"))

d_fit_taxa <- d_fit_taxa %>%
  mutate(PAF_t = pnorm(EC, mean = mu_t, sd = SDV_t))

SSD_taxa <- ggplot()+
  #geom_point(data = ecotox_data_TAXA, aes(x = log.EC10, y = frac_t, colour
  =SpeciesGroup_Harmonized)) +
  scale_color_manual(values = cols)+
  geom_line(data = d_fit_taxa,
    aes(x = EC,
      y = PAF_t,
      color = SpeciesGroup_Harmonized,
      Size = 1)) +
  labs(x = expression(paste("log10 [EC10 (µg/L)]")),
    y = "Potentially affected species fraction")+

  facet_wrap(~Name, ncol = 1, labeller = new_label_full_split ) +
  theme(strip.text.x = element_text(size = 20))

SSD_taxa

####save output with points

SSD_taxa_points <- ggplot()+
  geom_point(data = ecotox_data_TAXA, aes(x = log.EC10, y = frac_t, colour
  =SpeciesGroup_Harmonized), size = 5) +
  scale_color_manual(values = cols)+
  geom_line(data = d_fit_taxa,
    aes(x = EC,
      y = PAF_t,
      color = SpeciesGroup_Harmonized,
      Size = 2)) +

```

```

labs(x = expression(paste('log10 [EC10 (µg/L)]')),
     y = 'Potentially affected species fraction')+

facet_wrap(~Name, ncol=1,labeller = new_label_full_split ) +
theme(strip.text.x = element_text(size = 20))

save_and_display(
  SSD_taxa_points,
  "Herbicide_1-8.single SSD_no CI_no split",
  width = 7,
  height = 18)

SSD_taxa_points

### Adding CI to chemicals deterministic curve
#for herbicides newxs <- seq(0.001, 100, length.out = 1000)
#for insecticides newxs <- seq(-0.1, 10, length.out = 1000)
#insectides requires much lower starting scales 0.001

d.in <- ecotox_data_CAS
myboot2 <- function(d.in, newxs){
  xr <- sample(d.in$log.EC10, length(d.in$log.EC10), replace = TRUE)

  # fit distribution to new data
  fitr <- fitdistr(xr, 'normal')

  ##predict PAF for new data
  pyr <- pnorm(newxs, mean = fitr$estimate[1], sd = fitr$estimate[2])
  return(pyr)
}

pdat.final <- NULL
bootdat.final <- NULL
set.of.CAS_Harmonized <- unique(d.in$CAS_Harmonized)
id <- set.of.CAS_Harmonized[1]

for(id in set.of.CAS_Harmonized){
  data.sub <- d.in[which(d.in$CAS_Harmonized %in% id),]

  boots <- replicate(1000, myboot2(d.in=data.sub, newxs))

  # extract bootstrap values
  bootdat <- data.frame(boots)
  bootdat$newxs <- newxs
  bootdat <- reshape2::melt(bootdat, id = 'newxs')
  # extract CI
  cis <- apply(boots, 1, quantile, c(0.025, 0.975))
  rownames(cis) <- c('lwr', 'upr')

```



```

# add fitted values
pdat <- data.frame(newxs, py = pnorm(newxs, mean = data.sub$mu_C, sd =
data.sub$SDV_C))
# add CI
pdat <- cbind(pdat, t(cis))
pdat$CAS_Harmonized <- id

bootdat$CAS_Harmonized <- id

pdat.final <- rbind(pdat.final, pdat)
bootdat.final <- rbind(bootdat.final, bootdat)

}

#### Chemical CI

SSD_CAS_CI <- SSD_chemical

for(id in set.of.CAS_Harmonized){
  print(id)

  SSD_CAS_CI <- SSD_CAS_CI +

  geom_ribbon(data = pdat.final[which(pdat.final$CAS_Harmonized %in% id),], aes(x
=newxs, ymin =lwr,ymax = upr),
    fill="black", alpha=0.2)

}

SSD_CAS_CI

##Add original data
SSD_chemical2 <- SSD_CAS_CI +
  geom_segment(aes(x =ecotox_data_CAS$HC20_C, y = 0.0, xend =
ecotox_data_CAS$HC20_C, yend = 0.2),color ="red", size=0.9) +

  geom_segment(aes(x =ecotox_data_CAS$HC5_C, y = 0.0, xend =
ecotox_data_CAS$HC5_C, yend = 0.05),color ="black", size=0.9) +
  geom_point(data = ecotox_data_CAS, aes(x = log.EC10, y = frac_C, colour
=SpeciesGroup_Harmonized, shape=SpeciesGroup_Harmonized),size=5) +
  annotation_logticks(sides='b')+
  theme(legend.key.size = unit(0.5, 'cm'),
    legend.key.height = unit(0.5, 'cm'),
    legend.key.width = unit(0.5, 'cm'),
    legend.title = element_text(size=20),
    legend.text = element_text(size=20))+
  geom_errorbarh(data=ecotox_data_CAS, mapping=aes(y=frac_C, x=log.EC10,
xmin=log10(min_c), xmax=log10(max_c), colour = SpeciesGroup_Harmonized),
height=0.02, size=0.4)+
  geom_text(data=ecotox_data_CAS, aes(-1, 0.9,label=labels), size=5.5)+

```

```

scale_x_continuous(limits = c(-2,6))

SSD_chemical2

#### CI around taxonomic groups fitted curves

d.in <- ecotox_data_TAXA
myboot2 <- function(d.in, newxs){
  xr <- sample(d.in$log.EC10, length(d.in$log.EC10), replace = TRUE)

  # fit distribution to new data
  fitr <- fitdistr(xr, 'normal')

  ##predict PAF for new data
  pyr <- pnorm(newxs, mean = fitr$estimate[1], sd = fitr$estimate[2])
  return(pyr)
}

pdat.final <- NULL
bootdat.final <- NULL
set.of.CAS_Harmonized <- unique(d.in$CAS_Harmonized)
id <- set.of.CAS_Harmonized[1]

set.of.groups <- (unique(d.in$SpeciesGroup_Harmonized))
input.group <- set.of.groups
input.group
input.group <- set.of.groups[2]

for(id in set.of.CAS_Harmonized){
  data.sub <- d.in[which(d.in$CAS_Harmonized %in% id),]

  for(input.group in set.of.groups){
    data.sub.in <- data.sub[which(data.sub$SpeciesGroup_Harmonized %in% input.group),]

    boots <- replicate(1000, myboot2(d.in=data.sub.in, newxs))

    # extract bootstrap values
    bootdat <- data.frame(boots)
    bootdat$newxs <- newxs
    bootdat <- reshape2::melt(bootdat, id = 'newxs')
    # extract CI
    cis <- apply(boots, 1, quantile, c(0.025, 0.975))
    rownames(cis) <- c('lwr', 'upr')
    # add fitted values
    pdat <- data.frame(newxs, py = pnorm(newxs, mean = data.sub.in$mu_t, sd =
data.sub.in$SDV_t))
    # add CI
    pdat <- cbind(pdat, t(cis))
  }
}

```

```

pdat$CAS_Harmonized <- id
pdat$SpeciesGroup_Harmonized <- input.group

bootdat$CAS_Harmonized <- id
bootdat$SpeciesGroup_Harmonized <- input.group

pdat.final <- rbind(pdat.final, pdat)
bootdat.final <- rbind(bootdat.final, bootdat)

}
}

####Species group CI

SSD_taxa_CI <- SSD_taxa
for(id in set.of.CAS_Harmonized){
  print(id)
  for(input.group in set.of.groups){
    print(input.group)

    SSD_taxa_CI <- SSD_taxa_CI +

      geom_ribbon(data = pdat.final[which(pdat.final$CAS_Harmonized %in% id &
pdat.final$SpeciesGroup_Harmonized %in% input.group),], aes(x =newxs, ymin =lwr,ymax
= upr),
        fill="black", alpha=0.2)

  }
}
SSD_taxa_CI

####Observation per species groups
ecotox_data_TAXA$reduced_species.group.name <-
ifelse(ecotox_data_TAXA$SpeciesGroup_Harmonized=="Invertebrates", "I",

ifelse(ecotox_data_TAXA$SpeciesGroup_Harmonized=="Vertebrates", "V",

      "A"))

ecotox_data_TAXA$labels.taxa <- paste0(ecotox_data_TAXA$reduced_species.group.name,
" = ", ecotox_data_TAXA$n_t)

##position of the text
ecotox_data_TAXA$y_position <-
as.numeric(factor(ecotox_data_TAXA$reduced_species.group.name)) *0.05 + 0.80#0.65

####Add original data
SSD_taxa2 <- SSD_taxa_CI +

```

```

geom_segment(aes(x = ecotox_data_TAXA$HC20_t, y = 0.0, xend =
ecotox_data_TAXA$HC20_t, yend = 0.2), color = "red", size = 0.9) +

geom_segment(aes(x = ecotox_data_TAXA$HC5_t, y = 0.0, xend =
ecotox_data_TAXA$HC5_t, yend = 0.05), color = "black", size = 0.9) +
geom_point(data = ecotox_data_TAXA, aes(x = log.EC10, y = frac_t, colour
= SpeciesGroup_Harmonized, shape = SpeciesGroup_Harmonized), size = 5) +
scale_color_manual(values = cols) +
annotation_logticks(sides = "b") +
theme(legend.key.size = unit(0.5, "cm"),
      legend.key.height = unit(0.5, "cm"),
      legend.key.width = unit(0.5, "cm"),
      legend.title = element_text(size = 15),
      legend.text = element_text(size = 15)) +
geom_errorbarh(data = ecotox_data_TAXA, mapping = aes(y = frac_t, x = log.EC10,
xmin = log10(min_c), xmax = log10(max_c), colour = SpeciesGroup_Harmonized),
height = 0.02, size = 0.4) +
geom_text(aes(-0.7, ecotox_data_TAXA$y_position, label = ecotox_data_TAXA$labels.taxa),
size = 5.5, family = "Sans", fontface = "plain") +
scale_x_continuous(limits = c(-2, 6))

```

```

### Grid arrange first split plots
# Here all species groups are plotted separately

```

```

p <- plot_grid(SSD_chemical2 + theme(legend.position = "none", axis.text =
element_text(size = 15), axis.title = element_text(size = 17)),
               SSD_taxa2 + theme(legend.position = "right", axis.text = element_text(size =
15), axis.title = element_text(size = 17)),

```

```

      labels = c(""),
      label_x = 0.4,
      ncol = 2,
      rel_widths = c(1.4, 2.0),
      rel_heights = 10,
      align = "h",
      label_size = 25)

```

```

## Second splitting
# calculating PAF after combining vertebrates and other plants species groups
# This is where we combined data points for various groups based on the plot output
# Always check the species that need to be split and change color scheme accordingly

```

```

ecotox_data_rich$Merged_group <-
ifelse(ecotox_data_rich$SpeciesGroup_Harmonized == "Algae, cyanobacteria and aquatic
plants", "Algae, cyanobacteria and aquatic plants", "Others")

```

```

ecotox_data_rich %>%
  group_by(CAS_Harmonized, Merged_group) %>%
  arrange((log.EC10), .by_group = TRUE) %>%
  mutate(frac_t = ppoints(log.EC10, 0.5)) %>%

```

```

ungroup()->ecotox_data_rich_taxa.2

####New effect factors
ecotox_data_rich_taxa.2 <-ecotox_data_rich_taxa.2 %>%
  group_by(CAS_Harmonized, Merged_group) %>%
  mutate(n_t = n(),
         mu_t = mean(log.EC10),
         SDV_t = sd(log.EC10)) %>%

ungroup()

####Observation per merged species groups
ecotox_data_rich_taxa.2$reduced_species.group.name.2 <-
ifelse(ecotox_data_rich_taxa.2$Merged_group=="Algae, cyanobacteria and aquatic plants",
"A",

"Other")

ecotox_data_rich_taxa.2$labels.taxa.2 <-
paste0(ecotox_data_rich_taxa.2$reduced_species.group.name.2, " = ",
ecotox_data_rich_taxa.2$n_t)

ecotox_data_rich_taxa.2

##position of the text
ecotox_data_rich_taxa.2$y_position <-
as.numeric(factor(ecotox_data_rich_taxa.2$reduced_species.group.name.2))*0.05 + 0.85

ecotox_data_rich_taxa.2 <- ecotox_data_rich_taxa.2%>%
  mutate(HC20_t = qnorm(0.2, mean=mu_t, sd=SDV_t),
         HC5_t = qnorm(0.05, mean=mu_t, sd=SDV_t))

### Merged group deterministic curve
#Change the color depending on the grouping selected

d_fit_taxa.2 <- unique(ecotox_data_rich_taxa.2[c("CAS_Harmonized")])
d_fit_taxa.2 <- as.data.frame(d_fit_taxa.2)
colnames(d_fit_taxa.2) [1] <- c("CAS_Harmonized")

distribution_prameters_t <- ecotox_data_rich_taxa.2 %>%
  group_by(CAS_Harmonized, Merged_group) %>%
  slice(1)
####facet_grid new labels (manually done)
new_label <- as_labeller(c(`Simazine` = "Responsible split"))

d_fit_taxa.2 <- d_fit_taxa.2 %>%
  left_join(distribution_prameters_t[,c( "CAS_Harmonized","Merged_group",
"mu_t","SDV_t", "Name")],

```

```

      by=c("CAS_Harmonized")) %>%
right_join(
  tidy::expand(d_fit_taxa.2, CAS_Harmonized, EC = newxs),
  by = c("CAS_Harmonized"))

d_fit_taxa.2 <- d_fit_taxa.2 %>%
  mutate(PAF_t=pnorm(EC, mean=mu_t, sd=SDV_t))

SSD_taxa.2 <- ggplot()+
  #geom_point(data = ecotox_data_rich_taxa.2, aes(x = log.EC10, y = frac_t, colour
=Merged_group)) +
  scale_color_manual(values = cols.3)+
  geom_line(data=d_fit_taxa.2,
    aes(x=EC,
      y=PAF_t,
      color= Merged_group,
      Size=2)) +
  # scale_x_continuous(limits = c(0.010, 10),
  #   trans = log10_trans()) +

  labs(x = expression(paste('log10 [EC10 (µg/L)]')),
    y = 'Potentially affected species fraction')+
  #facet_grid(cols=vars(Name))+
  facet_wrap(~Name, ncol=1, labeller=new_label) +
  theme(strip.text.x = element_text(size = 20))

SSD_taxa.2

##### CI around taxonomic groups fitted curves
###combined groups after splitting

d.in <-ecotox_data_rich_taxa.2
myboot2 <- function(d.in, newxs){
  xr <- sample(d.in$log.EC10, length(d.in$log.EC10), replace = TRUE)

  # fit distribution to new data
  fitr <- fitdistr(xr, 'normal')

  ##predict PAF for new data
  pyr <- pnorm(newxs, mean = fitr$estimate[1], sd = fitr$estimate[2])
  return(pyr)
}

pdat.final <- NULL
bootdat.final <- NULL
set.of.CAS_Harmonized <- unique(d.in$CAS_Harmonized)
id <- set.of.CAS_Harmonized[1]

set.of.groups <- (unique(d.in$Merged_group))

```

```

input.group <- set.of.groups
input.group
input.group <- set.of.groups[2]

for(id in set.of.CAS_Harmonized){
  data.sub <- d.in[which(d.in$CAS_Harmonized %in% id),]
  #set.of.groups <- as.character(unique(data.sub$SpeciesGroup_Harmonized))

  for(input.group in set.of.groups){
    data.sub.in <- data.sub[which(data.sub$Merged_group %in% input.group),]

    # new data to predict
    #newxs <- seq(0.01, max(data.sub.in$log.EC10), length.out = 1000)
    boots <- replicate(1000, myboot2(d.in=data.sub.in, newxs))

    # extract bootstrap values
    bootdat <- data.frame(boots)
    bootdat$newxs <- newxs
    bootdat <- reshape2::melt(bootdat, id = 'newxs')
    # extract CI
    cis <- apply(boots, 1, quantile, c(0.025, 0.975))
    rownames(cis) <- c('lwr', 'upr')
    # add fitted values
    pdat <- data.frame(newxs, py = pnorm(newxs, mean = data.sub.in$mu_t, sd =
data.sub.in$SDV_t))
    # add CI
    pdat <- cbind(pdat, t(cis))
    pdat$CAS_Harmonized <- id
    pdat$Merged_group <- input.group

    bootdat$CAS_Harmonized <- id
    bootdat$Merged_group <- input.group

    pdat.final <- rbind(pdat.final, pdat)
    bootdat.final <- rbind(bootdat.final, bootdat)

  }
}
#### Add CI to the merged species groups
SSD_taxa_CI.2 <- SSD_taxa.2
for(id in set.of.CAS_Harmonized){
  print(id)
  for(input.group in set.of.groups){
    print(input.group)

    SSD_taxa_CI.2 <- SSD_taxa_CI.2 +

    geom_ribbon(data = pdat.final[which(pdat.final$CAS_Harmonized %in% id &
pdat.final$Merged_group %in% input.group),], aes(x = newxs, ymin = lwr, ymax = upr),

```

```

    fill="black", alpha=0.2)

  }
}

SSD_taxa.2. <- SSD_taxa_CI.2 +
  geom_segment(aes(x = ecotox_data_rich_taxa.2$HC20_t, y = 0.0, xend =
ecotox_data_rich_taxa.2$HC20_t, yend = 0.2), color = "red", size = 0.9) +
  geom_segment(aes(x = ecotox_data_rich_taxa.2$HC5_t, y = 0.0, xend =
ecotox_data_rich_taxa.2$HC5_t, yend = 0.05), color = "black", size = 0.9) +
  geom_point(data = ecotox_data_rich_taxa.2, aes(x = log.EC10, y = frac_t, colour
=Merged_group, shape = SpeciesGroup_Harmonized), size = 5) +
  scale_color_manual(values = cols.3) +
  annotation_logticks(sides = "b") +
  theme(legend.key.size = unit(0.5, 'cm'),
        legend.key.height = unit(0.5, 'cm'),
        legend.key.width = unit(0.5, 'cm'),
        legend.title = element_text(size = 15),
        legend.text = element_text(size = 15)) +

  geom_text(aes(-0.7,
ecotox_data_rich_taxa.2$y_position, label = ecotox_data_rich_taxa.2$labels.taxa.2), size = 5.5,
            family = "Sans", fontface = "plain", lineheight = .8) +
  geom_errorbarh(data = ecotox_data_rich_taxa.2, mapping = aes(y = frac_t, x = log.EC10,
xmin = log10(min_c), xmax = log10(max_c), colour = Merged_group), height = 0.02, size = 0.4) +

  scale_x_continuous(limits = c(-2, 6))

### Grid arrange all the three plots
SSD_taxa2 <- SSD_taxa2 + theme(
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank(),
  axis.title.y = element_blank())

SSD_taxa.2. <- SSD_taxa.2. + theme(
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank(),
  axis.title.y = element_blank())

merged.figure <- plot_grid(SSD_chemical2 + theme(legend.position = "none", axis.text =
element_text(size = 20), axis.title = element_text(size = 20)),
  SSD_taxa2 + theme(legend.position = "none", axis.text = element_text(size =
20), axis.title = element_text(size = 20)),
  SSD_taxa.2. + theme(legend.position = "none", axis.text = element_text(size
= 20), axis.title = element_text(size = 20)),
  labels = c("", ""),
  label_x = 0.4,
  ncol = 3,
  rel_widths = c(0.6, 0.6, 0.6), ##1.2
  rel_heights = 10,

```



```

      align = "h",
      label_size = 29)

title <- ggdraw() + draw_label("a.Simazine, CAS:122-34-9 (Photosynthesis inhibition
MoA):split algae, cyanobacteria and aquatic plants from the rest", fontface='bold', size=20)

merged.figure <- plot_grid(title, merged.figure, ncol=1, rel_heights=c(0.1, 1)) # rel_heights
values control title margins

save_and_display(
  merged.figure,
  "Split_ssds_Simazine",
  width = 21,
  height = 7)

```

References

- (1) Venables, W. N.; Ripley, B. D. *Modern Applied Statistics with S*, Fourth.; Springer: New York, 2002.
- (2) Emara, Y.; Joliet, O.; Finkbeiner, M.; Heß, S.; Kosnik, M.; Siegert, M.-W.; Fantke, P. Comparative Selective Pressure Potential of Antibiotics in the Environment. *Environ. Pollut.* **2023**, *318* (October 2022), 120873.
<https://doi.org/10.1016/j.envpol.2022.120873>.
- (3) Rosenbaum, R. K.; Georgiadis, S.; Fantke, P. Uncertainty Management and Sensitivity Analysis. In *Life Cycle Assessment: Theory and Practice*; Hauschild, M. Z., Rosenbaum, R. K., Olsen, S. I., Eds.; Springer International Publishing: Cham, 2018; pp 271–321. https://doi.org/10.1007/978-3-319-56475-3_11.

Excel Document with 180 underlying data and splitting SSD statistical results are provided as a supplementary Excel file.